Cheat Sheet

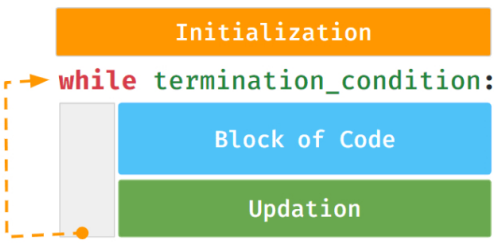
Loops

So far we have seen that Python executes code in a sequence and each block of code is executed once.

Loops allow us to execute a block of code several times.

While Loop

Allows us to execute a block of code several times as long as the condition is `True` .



While Loop Example

The following code snippet prints the next three consecutive numbers after a given number.

Code

PYTHON

```python
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
```

Input

```
4
```

Output

```
5
6
7
```

Possible Mistakes

1. Missing Initialization

Code

PYTHON

```python
a = int(input())
while counter < 3:
    a = a + 1
    print(a)
    counter = counter + 1
print("End")
```

Input

```
5
```

Output

```
NameError: name 'counter' is not defined
```

2. Incorrect Termination Condition

Code

PYTHON

```python
a = int(input())
counter = 0
condition = (counter < 3)
while condition:
    a = a + 1
    print(a)
    counter = counter + 1
```

Input

```
10
```

Output

The above code runs into an infinite loop.

While block will keep repeating as the value in condition variable is `True` .

3. Not Updating Counter Variable

Code

PYTHON

```python
a = int(input())
counter = 0
while counter < 3:
    a = a + 1
    print(a)
print("End")
```

Input

```
10
```

Output

```
Infinite Loop
```

As the value of counter is not updating.

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